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Data Analytics:

The DEVIL Lies In DATA

The world is facing an energy dilemma, and the only way out of it is to cut energy consumption in half. So how do we fill this demand-supply gap? Nikhil Pathak, vice president and general manager, IT business and SAARC, Schneider Electric India, speaks with Dilin Anand from EFY

Q. Connected devices existed before. What makes automation any different now?

A. While connected products have always existed, the first hurdle was to be able to get data. Once we developed the capability to get that, the problem of whether we could analyse it to glean information out of it came up. The most important thing about data analytics is that it gives the user the ability to predict.

Q. How does automation help improve efficiency in data centres?

A. Connectivity and software can help make energy more reliable and safe. This is done by using automation so as to minimise the number of waste cycles in any system or machinery. Critical cooling, server racks, integrated building management system and security usually drive data-centre infrastructure. These have elements, which when automated with an intelligent system could result in the deployment of proactive decisions that continuously keep the system running in its most efficient state.

Q. What is the importance of data here?

A. The devil lies in data. Data analytics lets us predict how a particular piece of equipment would behave. To simplify it, let us

consider that there is a battery in the adjacent room. We may assume it is getting charged, but it could be discharging right now, or even undergoing over-charge beyond its capacity. It could be heating up beyond its certified limits, which makes it prone to explosions. Instead of coming to know when the device eventually fails, we can be aware of the problem even before it happens.

Q. What is one ripe area for disruption in the server space?

A. Cooling aided by data could be a disruptor because it has a multiplier effect—if you get a chip that could withstand a few degrees higher, the savings you make on cooling systems could be enormous. As the heat generated by individual servers goes down, cumulative heat faced by the data centre would decrease. Sensors collecting temperature data allow cooling systems within servers to relax; power consumed by these would also go down. This way the cooling systems do not have to blow at full speed all the time.

Energy can be reduced further by setting fans to operate at less than full capacity without compromising performance at full load; under Fan Affinity Law, a fan working at up to 80 per cent capacity, for example, reduces energy

consumption by up to 50 per cent for the same performance.

Q. What is next for cooling?

A. Efficiency for refrigerants has led to their evolution, and today we have very efficient solutions available to us. Free cooling is the next-generation technology where you can cool the air without an AC, by mixing internal server air with ambient air available outside.

Software-controlled dimming and light temperature controls are also very popular these days.

Q. How else can we make a data centre more efficient?

A. Uninterrupted power supplies (UPSes) are a very important element here. Continuous and clean power would help data centres run more reliably. Modern UPSes have e-conversion, which can deliver up to 98 per cent efficiency using four-step inverter technology.

You also see active floors being used, which have sensors embedded into these to keep track of the environment.

Inverter-driven cooling is another important area where you see compressors and brushless DC motors being used a lot. A significant area of focus is in imbibing modularity into the systems so that scaling up becomes much easier. **EFY**